

$$(211)R(21) = 4 (11R1)$$

13. Answer: D

When two regions A and B are in thermal equilibrium, they exchange thermal energy at the same rate i.e A emits thermal energy to B and absorbs thermal energy from B at the same rate (there is no net heat transfer between A and B). When this happens, A and B are said to be of equal temperature.

X: input of thermal energy is more than the output of thermal energy (length of arrow towards X is longer than the arrow away from X), hence the initial temperature of X is below 30°C.

Y: length of the two arrows is the same, hence initial temperature of Y is same as the surrounding, 30°C.

Z: length of arrow towards Z is shorter than the arrow away from Z, hence the initial temperature of Z is higher than 30°C.

14. Answer: C